

Sequential Genetic Search for Ensemble Feature Selection

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Abstract

Ensemble learning constitutes one of the main directions in machine learning and data mining. Ensembles allow us to achieve higher accuracy, which is often not achievable with single models. One technique, which proved to be effective for constructing an ensemble of diverse classifiers, is the use of feature subsets. Among different approaches to ensemble feature selection, genetic search was shown to perform best in many domains. In this paper, a new strategy GAS-SEFS, Genetic Algorithm-based Sequential Search for Ensemble Feature Selection, is introduced. Instead of one genetic process, it employs a series of processes, the goal of each of which is to build one base classifier. Experiments on 21 data sets are conducted, comparing the new strategy with a previously considered genetic strategy for different ensemble sizes and for five different ensemble integration methods. The experiments show that GAS-SEFS, although being more time-consuming, often builds better ensembles, especially on data sets with larger numbers of features.

1 Introduction

A popular method for creating an accurate model from a set of training data is to construct a set (ensemble) of classifiers. It was shown that an ensemble is often more accurate than any of the single classifiers in it. The integration of classifiers is currently an active research area in the machine learning and neural networks communities [Dietterich, 1997].

Both theoretical and empirical research have demonstrated that a good ensemble should include diverse base classifiers. Another important issue in creating an effective ensemble is the choice of the function for combining the predictions of the base classifiers. It was shown that increasing the ensemble diversity is not enough to ensure increased accuracy – if the integration method does not properly utilize the ensemble diversity, then no benefit arises from the integration [Brodley and Lane, 1996].

One effective approach for generating an ensemble of diverse classifiers is the use of feature subsets, or *ensemble feature selection* [Opitz, 1999]. By varying the feature sub-

sets used to generate the base classifiers, it is possible to promote diversity and produce base classifiers that tend to err in different sub-areas of the instance space.

Feature selection algorithms, including ensemble feature selection, are typically composed of the following components [Aha and Bankert, 1995, Opitz, 1999]: (1) *search strategy*, that searches through the space of feature subsets; and (2) *fitness function*, that inputs a feature subset and outputs a numeric evaluation. The search strategy's goal is to find a feature subset maximizing this function.

It is reasonable to include in the fitness function, explicitly or implicitly, both accuracy and diversity. One measure of fitness, which was proposed in [Opitz, 1999], defines fitness $Fitness_i$ of classifier i corresponding to feature subset i to be proportional to classification accuracy acc_i and diversity div_i of the classifier:

$$Fitness_i = acc_i + \alpha \cdot div_i, \quad (1)$$

where α reflects the influence of diversity. Diversity div_i is the contribution of classifier i to the total ensemble diversity, which can be measured as the average pairwise diversity for all the pairs of classifiers including i . This fitness function was also used in experiments in [Tsymbal *et al.*, 2003; 2005], and it is used in the experiments in this paper.

In [Tsymbal *et al.*, 2005] a genetic search-based strategy GA has been introduced. It uses genetic search for evolving the initial population built with random subsampling. GA was shown to perform best on average with respect to the other three strategies, and two diversity measures were best for GA of the five considered measures: the kappa statistic, and the fail/non-fail disagreement.

In this paper, we introduce a new genetic search-based strategy for ensemble feature selection, GAS-SEFS, which, instead of maintaining a set of feature subsets in each generation like in GA, consists in applying a series of genetic processes, one for each base classifier, sequentially.

The paper is organized as follows. In Section 2 the task of ensemble feature selection is considered. In Section 3 we present two strategies for genetic ensemble feature selection, GA and GAS-SEFS, and two diversity measures. In Section 4 different methods for ensemble integration are reviewed. In Section 5 we present our experiments with the two genetic strategies and conclude in the next section with a summary and assessment of further research topics.

2 Ensemble Feature Selection and Random Subspacing

The task of using an ensemble of models can be broken down into two basic questions: (1) what set of models should be generated?; and (2) how should the predictions of the models be integrated? [Dietterich, 1997].

One effective approach to ensemble generation is the use of different subsets of features for each model. Finding a set of feature subsets for constructing an ensemble is also known as *ensemble feature selection* [Opitz, 1999]. While traditional feature selection algorithms have the goal of finding the best feature subset that is suitable to both the learning problem and the learning algorithm, the task of ensemble feature selection has the additional goal of finding a set of feature subsets that will promote diversity among the base classifiers [Opitz, 1999].

Ho [1998] has shown that simple random selection of features may be an effective technique for ensemble feature selection because the lack of accuracy in the ensemble members is compensated for by their diversity. This technique is called the random subspace method or simply Random Subspacing (RS).

With RS one may solve the small sample size problem, because the training sample size relatively increases in random subspaces. Ho [1998] shows that while most other classification methods suffer from the curse of dimensionality, this method does not. RS has much in common with bagging [Skurichina and Duin, 2001], but instead of sampling instances, one samples features. Like bagging, RS is a parallel learning algorithm, that is, generation of each base classifier is independent. This makes it suitable for parallel implementation that is desirable in some practical applications. It was shown that, like in bagging, accuracy could be only increased with the addition of new members, even when the ensemble complexity grew [Ho, 1998].

RS is used as a base in a number of ensemble feature selection strategies, e.g. GEFS (Genetic Ensemble Feature Selection) [Opitz, 1999] and HC (Hill Climbing) [Cunningham and Carney, 2000].

3 Two Strategies for Genetic Ensemble Feature Selection

3.1 GA and GAS-SEFS

The use of genetic search has been an important direction in the feature selection research. Genetic algorithms have been shown to be effective global optimization techniques. The use of genetic algorithms for ensemble feature selection was first proposed in [Kuncheva, 1993] and further elaborated in [Kuncheva and Jain, 2000]. As the fitness function in [Kuncheva, 1993; Kuncheva and Jain, 2000] the ensemble accuracy was used instead of the accuracy of the base classifiers. However, such a fitness function is biased towards some particular integration method (often simple voting). Besides, as it was shown e.g. in [Kuncheva, 1993], such a design is prone to overfitting, and some additional preventive measures are needed to be taken to avoid this (as in-

cluding in the fitness function penalty terms accounting for the number of features). The use of individual accuracy and diversity as in (1) is an alternative solution to this problem. Another motivation for this alternative is the fact that overfitting at the level of the base classifiers is more desirable than overfitting of the ensemble itself. It was shown recently in several studies that an ensemble of overfitted members might often be better than an ensemble of non-overfitted members. For example, in [Street and Kim, 2001] pruning trees resulted in decreased ensemble accuracy, even though the accuracy of the trees themselves increased.

The Genetic Algorithm for ensemble feature selection (GA) [Tsybal et al., 2005] is based on the GEFS strategy of Opitz [1999]. GEFS was the first genetic algorithm for ensemble feature selection that explicitly used diversity in its fitness function. GA begins with creating an initial population with RS. Then, new candidate classifiers are produced by crossover and mutation. After producing a certain number of individuals the process continues with selecting a new subset of candidates randomly with a probability proportional to fitness (so called roulette-wheel selection). The process of creating new classifiers and selecting a subset of them (a generation) continues a certain number of times. After a predefined number of generations, the fittest individuals make up the population, which comprises the ensemble. The representation of each individual is a constant-length string of bits, where each bit corresponds to a particular feature. The crossover operator uses uniform crossover, in which each feature of the two children takes randomly a value from one of the parents. The mutation operator randomly toggles a number of bits in an individual.

Instead of maintaining a set of feature subsets in each generation of one genetic process, GAS-SEFS (Genetic Algorithm-based Sequential Search for Ensemble Feature Selection) uses a series of genetic processes, one for each base classifier, sequentially. Pseudo-code for GAS-SEFS is given in Figure 1. After each genetic process one base classifier is selected into the ensemble. GAS-SEFS uses the same fitness function (1), but diversity is calculated with the base classifiers already formed by previous genetic processes instead of the members of current population. In the first GA process, the fitness function has to use accuracy only. GAS-SEFS uses the same genetic operators as GA.

```
For I=1 to EnsembleSize
  For J=1 to 10 Population(J)=RSM(#Features);
  For J=1 to #Generations
    For K=1 to 10 CalculateFitness(Population(K));
    For K=1 to 40
      //randomly proportional to log(1+fitness)
      (L,M)=Select2(Population);
      Offsprings(K)=CrossOver(L,M);
    EndForK
    For K=1 to 10 Mutate1_0(Offsprings(20+K));
      Mutate0_1(Offsprings(30+K));
    For K=1 to 40 CalculateFitness(Offsprings(K));
      //randomly proportional to fitness
      Population=Select10(Population+Offsprings);
    EndForJ
    //according to fitness
    BaseClassifier(I)=Select1(Population);
  EndForJ
```

Figure 1 Pseudo-code for the GAS-SEFS algorithm

GA has a number of peculiarities, which we use also in GAS-SEFS. Full feature sets are not allowed in RS nor may the crossover operator produce a full feature subset. Individuals for crossover are selected randomly proportional to $\log(1+fitness)$ instead of just fitness, which adds more diversity into the new population. The generation of children identical to their parents is prohibited. To provide a better diversity in the length of feature subsets, two different mutation operators are used (*Mutate1_0* and *Mutate0_1*), one of which always deletes features randomly with a given probability, and the other – adds features.

Parameter settings for our implementation of GA and GAS-SEFS include a mutation rate of 50%, a population size of 10, a search length of 40 feature subsets (the number of new individuals produced by crossover and mutation), of which 20 are offsprings of the current population of 10 classifiers generated by crossover, and 20 are mutated offsprings (10 with each mutation operator). 10 generations of individuals were produced, as our pilot studies have shown that in most cases, with this configuration, the ensemble accuracy does not improve after 10 generations, due to overfitting the training data.

The complexity of GA does not depend on the number of features, and is $O(S' \cdot N_{gen})$, where S' is the number of individuals in one generation, and N_{gen} is the number of generations [Tsymbal *et al.*, 2005]. The complexity of GAS-SEFS is $O(S \cdot S' \cdot N_{gen})$, where S is the number of base classifiers. In our experiments, on average, GA and GAS-SEFS look through about 400 and 4000 feature subsets correspondingly (given that the number of base classifiers is 10, the number of individuals in a generation is 40, and the number of generations is 10).

3.2 Diversity measures used in the fitness function

The fail/non-fail disagreement measure and the kappa statistic were shown to provide the best performance for GA in [Tsymbal *et al.*, 2005].

The *fail/non-fail disagreement* measure was defined in [Skalak, 1996] as the percentage of test instances for which the classifiers make different predictions but for which one of them is correct:

$$div_dis_{ij} = \frac{N^{01} + N^{10}}{N^{11} + N^{10} + N^{01} + N^{00}}, \quad (2)$$

where N^{ab} is the number of instances, classified correctly ($a=1$) or incorrectly ($a=0$) by classifier i , and correctly ($b=1$) or incorrectly ($b=0$) by classifier j . The denominator in (2) is equal to the total number of instances. The fail/non-fail disagreement varies from 0 to 1.

The *kappa* statistic was first introduced in [Cohen, 1960]. Let N_{ij} be the number of instances, recognized as class i by the first classifier and as class j by the second one, N_{i*} is the number of instances recognized as i by the first classifier, and N_{*j} is the number of instances recognized as j by the second classifier. Define then Θ_1 and Θ_2 as:

$$\Theta_1 = \frac{\sum_{i=1}^l N_{ii}}{N}, \text{ and } \Theta_2 = \sum_{i=1}^l \left(\frac{N_{i*}}{N} \cdot \frac{N_{*i}}{N} \right), \quad (3)$$

where l is the number of classes and N is the total number of instances. Θ_1 estimates the probability that the two classifiers agree, and Θ_2 is a correction term, which estimates the probability that the two classifiers agree simply by chance (in the case where each classifier chooses to assign a class label randomly). The pairwise diversity div_kappa_{ij} is defined as follows:

$$div_kappa_{ij} = \frac{\Theta_1 - \Theta_2}{1 - \Theta_2}. \quad (4)$$

We normalize this measure to vary from 0 to 1.

4 Integration of an Ensemble of Models

The challenging problem of integration is to decide which of the classifiers to select or how to combine the results produced by the base classifiers. A number of *selection* and *combination* approaches have been proposed.

One of the most popular and simplest techniques used to combine the results of the base classifiers, is simple voting (also called majority voting) [Bauer and Kohavi, 1999]. In the voting, the output of each base classifier is considered as a vote for that particular class value. The class value that receives the biggest number of votes is selected as the final classification. Weighted Voting (WV), where each vote has a weight proportional to the estimated generalization performance of the corresponding classifier, works usually better than simple voting [Bauer and Kohavi, 1999].

A number of selection techniques have also been proposed to solve the integration problem. One of the most popular and simplest selection techniques is Cross-Validation Majority (CVM, also called Single Best; we call it simply Static Selection, SS, in our experiments) [Schaffer, 1993]. In CVM, the cross-validation accuracy for each base classifier is estimated, and then the classifier with the highest accuracy is selected.

The described above approaches are *static*. They select one model for the whole data space or combine the models uniformly. In *dynamic* integration each new instance to be classified is taken into account. Usually, better results can be achieved if integration is dynamic.

We consider in our experiments three dynamic techniques based on the same local error estimates: Dynamic Selection (DS), Dynamic Voting (DV), and Dynamic Voting with Selection (DVS) [Tsymbal and Puuronen, 2000]. They contain two main phases. First, at the learning phase, the local classification errors of each base classifier for each instance of the training set are estimated according to the 1/0 loss function using cross validation. The learning phase finishes with training the base classifiers on the whole training set. The application phase begins with determining k -nearest neighbours for a new instance using a given distance metric. Then, weighted nearest neighbour regression is used to predict the local classification errors of each base classifier for the new instance.

After, DS simply selects a classifier with the least predicted local classification error. In DV, each base classifier receives a weight that is proportional to its estimated local accuracy, and the final classification is produced as in WV. In DVS, the base classifiers with the highest local classification errors are discarded (the classifiers with errors that fall into the upper half of the error interval) and locally weighted voting (DV) is applied to the remaining classifiers.

5 Experimental Investigations

5.1 Experimental setup

The experiments are conducted on 21 data sets taken from the UCI machine learning repository [Blake *et al.*, 1999]. These data sets include real-world and synthetic problems, vary in characteristics, and were previously investigated by other researchers. The main characteristics of the data sets are presented in Table 1.

Table 1 Data sets and their characteristics

Data set	Instances	Classes	Features	
			Categ.	Num.
Balance	625	3	0	4
Breast Cancer	286	2	9	0
Car	1728	4	6	0
Diabetes	768	2	0	8
Glass Recognition	214	6	0	9
Heart Disease	270	2	0	13
Ionosphere	351	2	0	34
Iris Plants	150	3	0	4
LED	300	10	7	0
LED17	300	10	24	0
Liver Disorders	345	2	0	6
Lymphography	148	4	15	3
MONK-1	432	2	6	0
MONK-2	432	2	6	0
MONK-3	432	2	6	0
Soybean	47	4	0	35
Thyroid	215	3	0	5
Tic-Tac-Toe	958	2	9	0
Vehicle	846	4	0	18
Voting	435	2	16	0
Zoo	101	7	16	0

As in [Tsymbal *et al.*, 2003; 2005], we use Simple Bayes (SB) as the base classifier in the ensembles. It has been recently shown experimentally and theoretically that SB can be optimal even when the “naïve” feature-independence assumption is violated by a wide margin [Domingos and Pazzani, 1997]. Second, when SB is applied to the sub-problems of lower dimensionalities, the error bias of the Bayesian probability estimates caused by the feature-independence assumption becomes smaller. It also can easily handle missing feature values. Besides, it has advantages in terms of simplicity, learning speed, classification speed, and storage space. We believe that dependencies and conclusions presented in this paper do not depend on the learning algorithm used and would be similar for most known learning algorithms.

To evaluate GA and GAS-SEFS, we have used stratified random-sampling cross validation with 60 percent of instances in the training set. The remaining 40 percent of in-

stances were divided into two sets of approximately equal size (a validation set and a test set). 70 test runs of were made on each data set for each search strategy and diversity.

Four different ensemble sizes have been tested: 3, 5, 7, and 10. The ensemble size did not exceed 10 due to two main reasons: (1) limitation in computational resources, and (2) it was shown in experiments that for guided ensemble construction such as genetic search the biggest gain is achieved already with 10 base classifiers, and much less classifiers are needed than with unguided ensemble construction such as RS and bagging.

At each run of the algorithm, accuracies for the five types of ensemble integration are collected: Static Selection (SS), Weighted Voting (WV), Dynamic Selection (DS), Dynamic Voting (DV), and Dynamic Voting with Selection (DVS). We have collected ensemble characteristics for four numbers of generations: 1, 3, 5, and 10.

To reduce the number of possible combinations of parameters, we conducted a separate series of preliminary experiments using the wrapper approach based on cross validation to select the best diversity coefficient α and the number of nearest neighbors in dynamic integration k as in [Tsymbal *et al.*, 2005]. We have experimented with seven values of α : 0, 0.25, 0.5, 1, 2, 4, and 8. Seven values: 1, 3, 7, 15, 31, 63, 127 ($2^n - 1$, $n = 1, \dots, 7$) were used for k . From the experimental results we could see that the best value of k depended mostly only on the integration method used and on the data set. The best α ’s varied with the search strategy, integration method, and data set used.

After, the experiments were repeated with the selected values of α and k . Although the same data were used for the selection and for the later experiments, we believe that this did not lead to overfitting due to the small number of possible values for α and k .

The test environment was implemented within the MLC++ framework (the machine learning library in C++) [Kohavi *et al.*, 1996]. A multiplicative factor of 1 was used for the Laplace correction in SB as in [Domingos and Pazzani, 1997]. Numeric features were discretized into ten equal-length intervals (or one per observed value, whichever was less), as it was done in [Domingos and Pazzani, 1997]. Although this approach was found to be slightly less accurate than more sophisticated ones, it has the advantage of simplicity, and is sufficient for comparing different ensembles of SB classifiers with each other.

5.2 Experimental results

To validate our findings, we divided all data sets into two groups: with less than 9 features (10 data sets, group 1), and with greater than or equal to 9 features (11 data sets, group 2); and checked all the characteristics for these groups. The ensemble accuracies were nearly the same for the two diversity measures, and the fail/non-fail disagreement was slightly better on average, so we present results for this diversity measure only here. In Figure 2 the ensemble accuracies for strategies GA and GAS-SEFS, over the two groups of data sets and four ensemble sizes are shown averaged over the data sets for the best integration method (DVS). It

can be seen from the figure that GAS-SEFS builds even more accurate ensembles than GA; especially for group 2 including data sets with larger numbers of features. Accuracy grows with the ensemble size, but the growth flattens as the number of base classifiers increases.

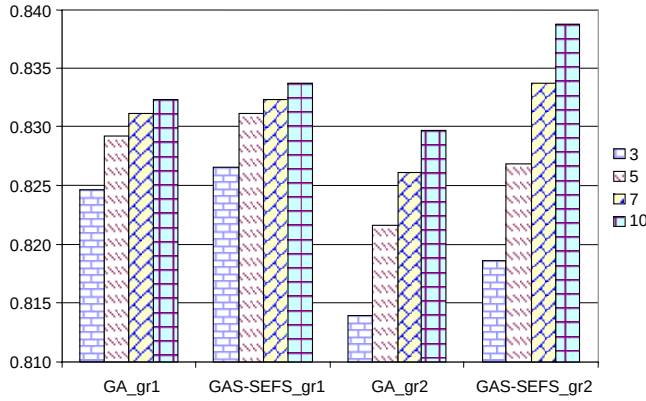


Figure 2 Ensemble accuracies for two strategies, two groups of data sets, and four ensemble sizes

In Figure 3 ensemble accuracies are shown for the two strategies, five integration methods, and four ensemble sizes on the Tic-Tac-Toe data set, as a representative of group 2 including 958 instances and 9 features. This figure supports our previous findings. Besides, it could be seen that dynamic integration, expectedly, outperforms static integration both for GA and for GAS-SEFS. Accuracy grows with the ensemble size and this growth is greater for the best integration methods (DS and DVS in this case).

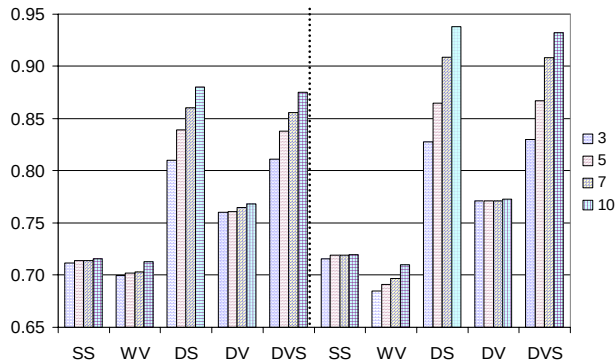


Figure 3 Ensemble accuracies for GA (left) and GAS-SEFS (right) for five integration methods and four ensemble sizes on the Tic-Tac-Toe data set

The difference between the two strategies is clearer for the best integration methods. The dependencies were the same for all the data sets, with sometimes lesser difference between the integration methods. For some data sets DV outperforms DS, which supports the previous findings about behaviour of the integration methods [Tsymbal and Puuronen, 2000; Tsymbal *et al.*, 2005].

5.3 Other interesting findings

Selected values of α were different for different data sets, supporting findings in [Tsymbal *et al.*, 2003; 2005]. In general, for both strategies, α for the dynamic integration methods is bigger than for the static ones (2.2 vs 0.8 on average). GAS-SEFS needs slightly higher values of α than GA (1.8 vs 1.5 on average). This can be explained by the fact that GAS-SEFS always starts with a classifier, which is based on accuracy only, and the subsequent classifiers need more diversity than accuracy.

The number of selected features falls as the ensemble size grows, and this is especially clear for GAS-SEFS, as the base classifiers need more diversity. As a rule, more features are needed in the static integration methods than in the dynamic ones to achieve better accuracy. GAS-SEFS results in slightly smaller feature subsets on average (48% vs 50% of features for dynamic integration strategies).

As it was also reported in [Tsymbal *et al.*, 2005], the selected k -neighbourhood values for dynamic integration change with the integration method. DS needs higher values of k . This can be explained by the fact that its prediction is based on only one classifier being selected, and thus, it is very unstable. Higher values of k provide more stability to DS. The average selected k is equal to 33 for DS, and it is only 14 for DV. For DVS, as a hybrid strategy, it is in between at 24. The selected values of k do not change significantly with the change of the search strategy and the ensemble size.

Experimental results for both GA and GAS-SEFS show that the static integration methods, SS and WV, and the dynamic DS start to overfit the validation set already after 5 generations and show lower accuracies, whereas the accuracies of DV and DVS continue to grow up to 10 generations. This shows the importance of selection of the appropriate integration method for the genetic strategies.

6 Conclusions

In our paper, we have considered two genetic search strategies for ensemble feature selection. The new strategy, GAS-SEFS, consists in employing a series of genetic search processes, one for each base classifier. It was shown in experiments that GAS-SEFS results in better ensembles having greater accuracy in many domains, especially for data sets with relatively larger numbers of features. GAS-SEFS is significantly more time-consuming than GA. However, it can be easily parallelized in a multiprocessor setting, and one processor could be used for each offspring in the current generation.

One of the reasons for the success of GAS-SEFS is the fact that each of the core GA processes leads to significant overfitting of a corresponding ensemble member, and, as it was shown before, an ensemble of overfitted members is often better than an ensemble of non-overfitted members.

In [Oliveira *et al.*, 2003] it was shown that besides the use of weights to combine a number of objectives in the fitness function in genetic algorithms (as the use of α in our case), another common approach that often gives better results for

single feature subset selection is based on Pareto-front dominating solutions. Adaptation of this technique to ensemble feature selection is an interesting topic for further research.

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